

WEEK 4 – Code Quality & Microservices with Springboot 3

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EXERCISE 1: Creating Microservices for account and loan

Task: In this hands on exercises, we will create two microservices for a bank. One microservice for handling accounts and one for handling loans. Each microservice will be a specific independent Spring RESTful Webservice maven project having it's own pom.xml. The only difference is that, instead of having both account and loan as a single application, it is split into two different applications. These webservices will be a simple service without any backend connectivity. Follow steps below to implement the two microservices:

Account Microservice

Loan Microservice

Step-by-Step Solution:

1. Account.java

```
package com.cognizant.account.model;
```

```
public class Account {
```

```
    private String number;
```

```
    private String type;
```

```
    private double balance;
```

```
    public Account() {
```

```
    }
```

```
    public Account(String number, String type, double balance) {
```

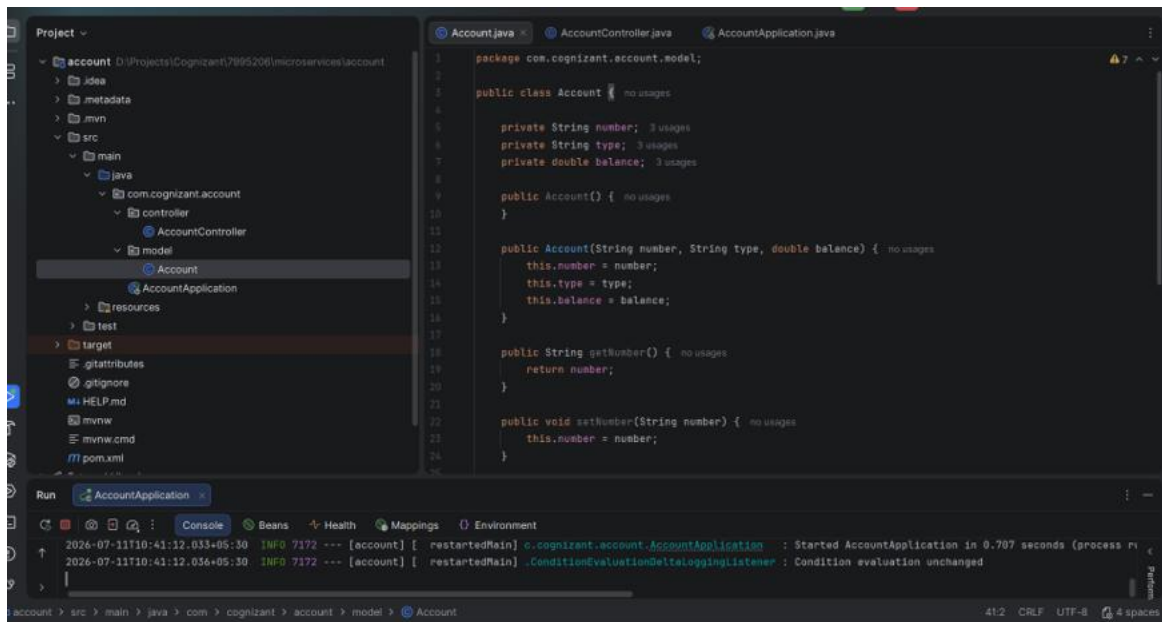
```
        this.number = number;
```

```
        this.type = type;
```

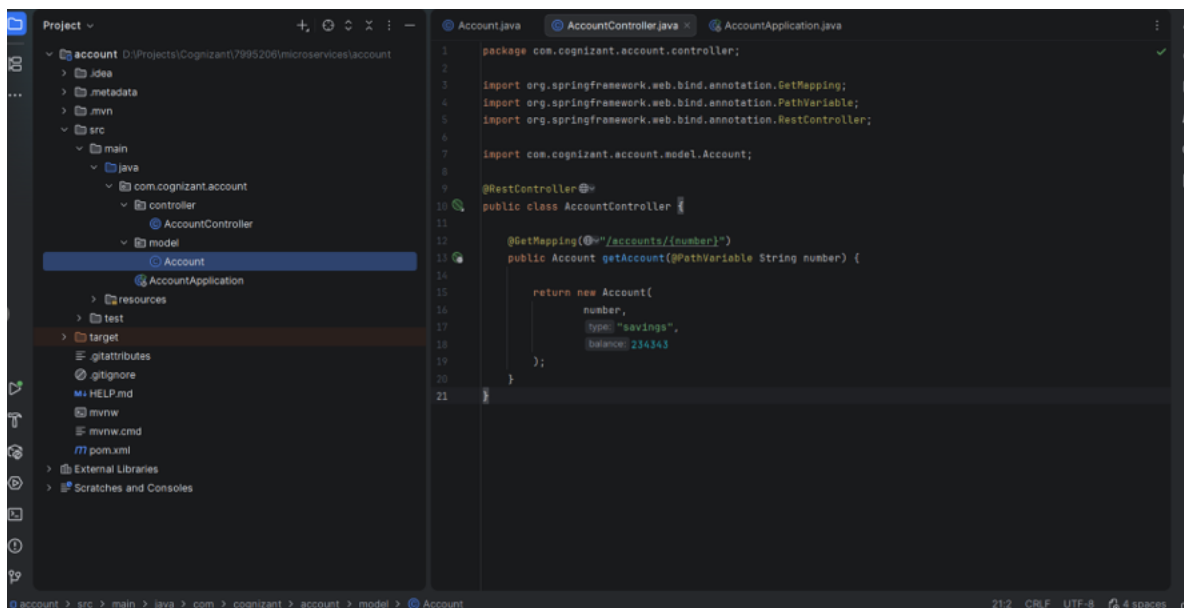
```
        this.balance = balance;
```

```
    }
```

```
public String getNumber() {  
    return number;  
}  
  
public void setNumber(String number) {  
    this.number = number;  
}  
  
public String getType() {  
    return type;  
}  
  
public void setType(String type) {  
    this.type = type;  
}  
  
public double getBalance() {  
    return balance;  
}  
  
public void setBalance(double balance) {  
    this.balance = balance;  
}  
}
```



2. AccountController.java



package com.cognizant.account.controller;

import org.springframework.web.bind.annotation.GetMapping;

import org.springframework.web.bind.annotation.PathVariable;

import org.springframework.web.bind.annotation.RestController;

import com.cognizant.account.model.Account;

```

@RestController
public class AccountController {

    @GetMapping("/accounts/{number}")
    public Account getAccount(@PathVariable String number) {

        return new Account(
            number,
            "savings",
            234343
        );
    }
}

```

Loan.java

```

package com.cognizant.loan.model;

public class Loan {

    private String number;
    private String type;
    private double loan;
    private double emi;
    private int tenure;

    public Loan() {
    }

    public Loan(String number, String type, double loan, double emi, int tenure) {
        this.number = number;
        this.type = type;
        this.loan = loan;
        this.emi = emi;
        this.tenure = tenure;
    }

    public String getNumber() {
        return number;
    }
}

```

```

    }

    public void setNumber(String number) {
        this.number = number;
    }

    public String getType() {
        return type;
    }

    public void setType(String type) {
        this.type = type;
    }

    public double getLoan() {
        return loan;
    }

    public void setLoan(double loan) {
        this.loan = loan;
    }

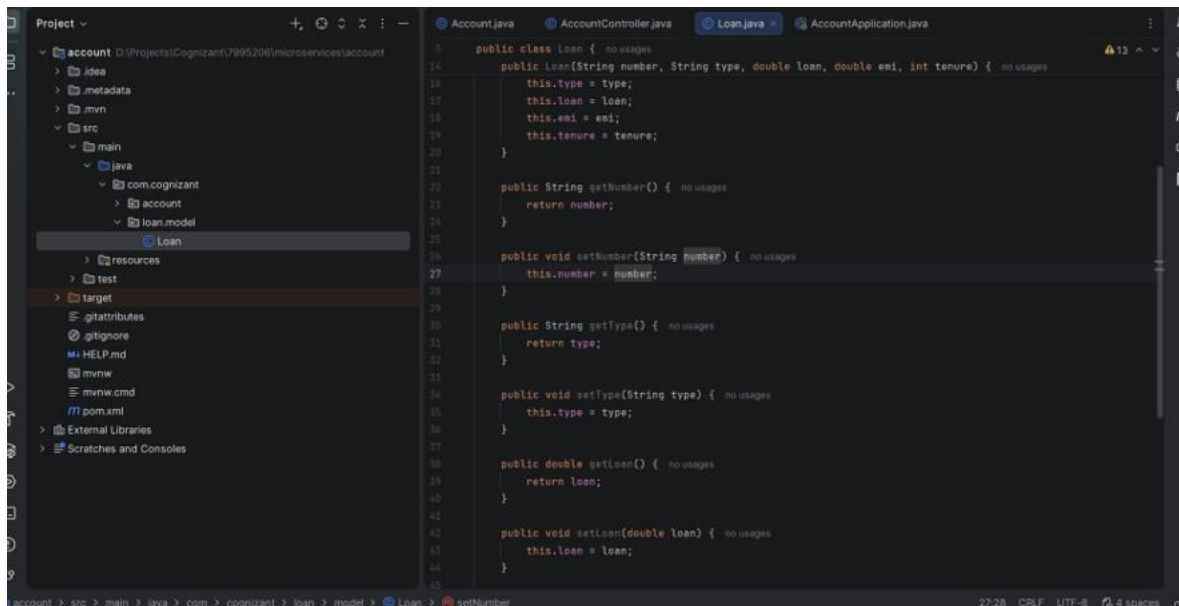
    public double getEmi() {
        return emi;
    }

    public void setEmi(double emi) {
        this.emi = emi;
    }

    public int getTenure() {
        return tenure;
    }

    public void setTenure(int tenure) {
        this.tenure = tenure;
    }
}

```



LoanController.java

```
package com.cognizant.loan.controller;
```

```
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
```

```
import com.cognizant.loan.model.Loan;
```

```
@RestController
```

```
public class LoanController {
```

```
    @GetMapping("/loans/{number}")
```

```
    public Loan getLoan(@PathVariable String number) {
```

```
        return new Loan(
```

```
            number,
```

```
            "car",
```

```
            400000,
```

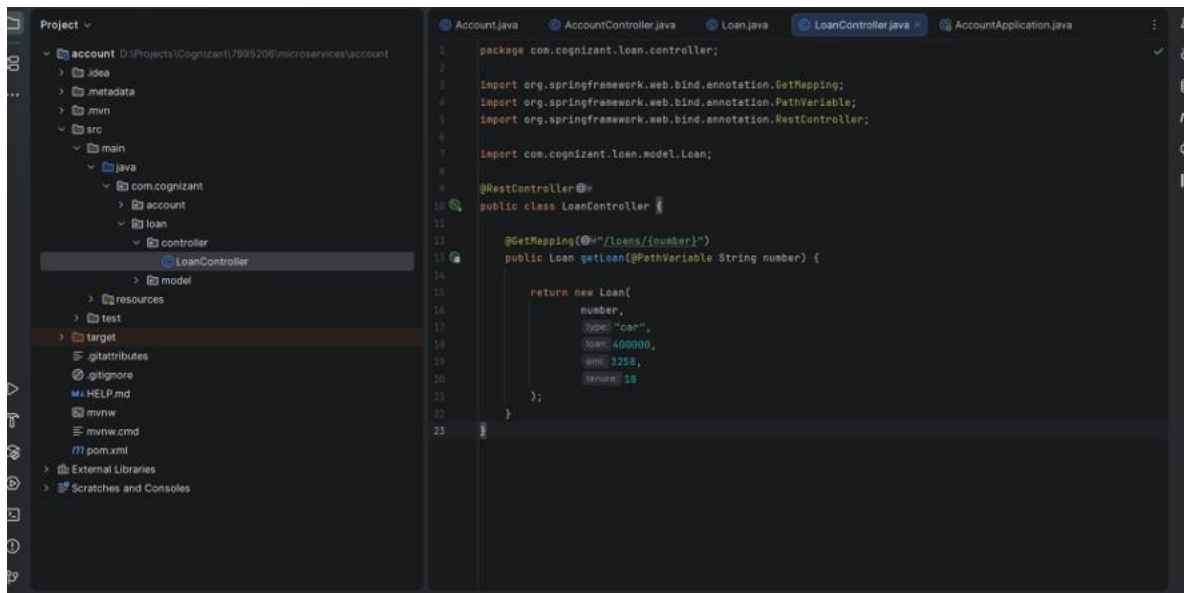
```
            3258,
```

```
            18
```

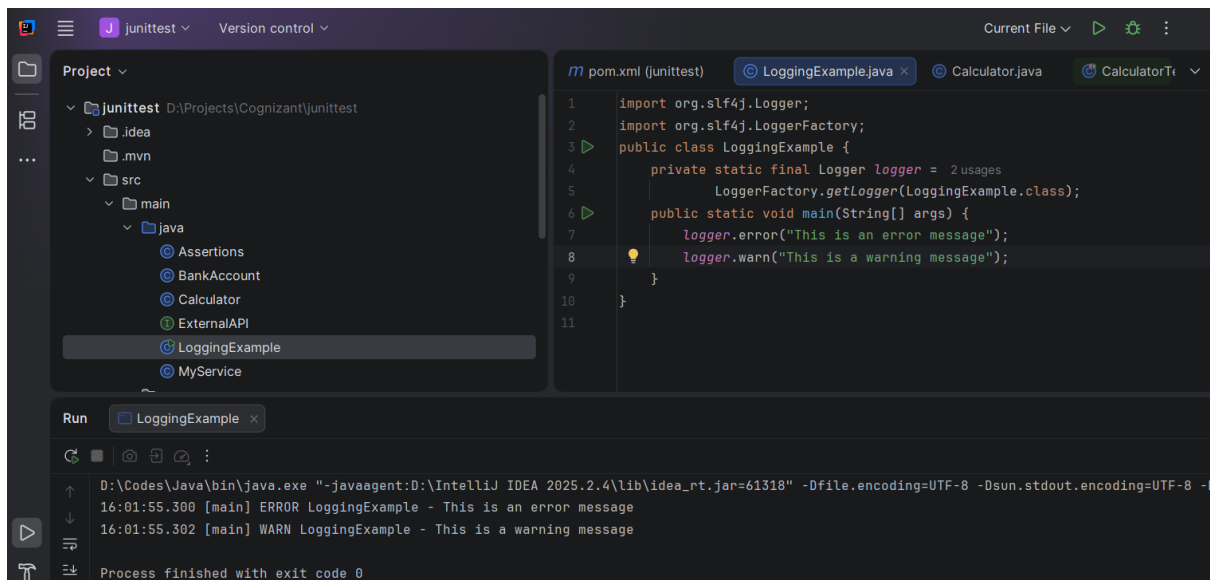
```
        );
```

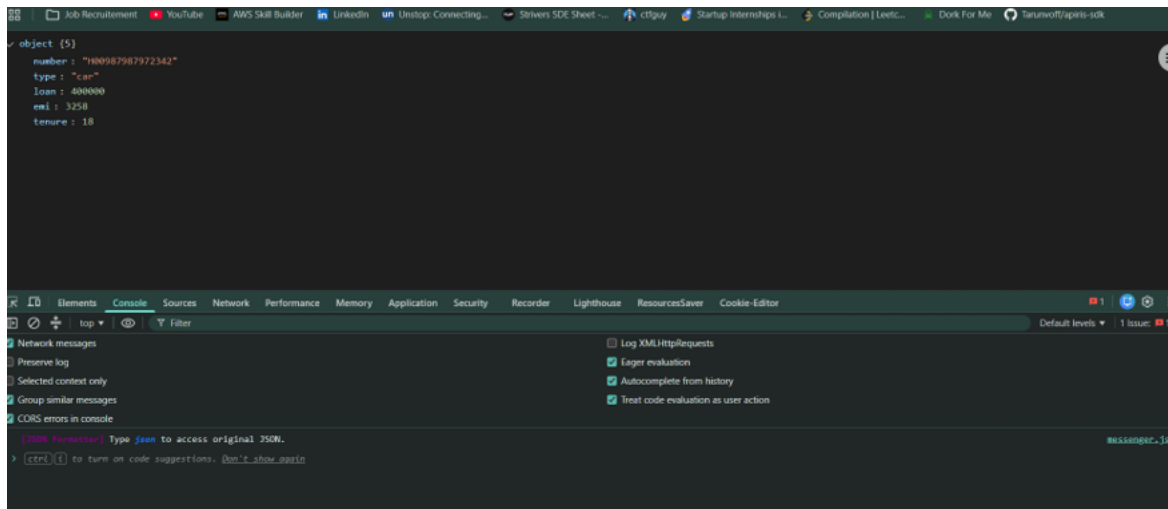
```
    }
```

```
}
```



OUTPUT





EXERCISE 2: Create Eureka Discovery Server and register

1. Account.java

```
package com.cognizant.account.model;
```

```
public class Account {
```

```
    private String number;
```

```
    private String type;
```

```
    private double balance;
```

```
    public Account() {
```

```
    }
```

```
    public Account(String number, String type, double balance) {
```

```
        this.number = number;
```

```
        this.type = type;
```

```
        this.balance = balance;
```

```
    }
```



```
public String getNumber() {  
    return number;  
}
```

```
public void setNumber(String number) {  
    this.number = number;  
}
```

```
public String getType() {  
    return type;  
}
```

```
public void setType(String type) {  
    this.type = type;  
}
```

```
public double getBalance() {  
    return balance;  
}
```

```
public void setBalance(double balance) {  
    this.balance = balance;  
}  
}
```

AccountController.java

```
package com.cognizant.account.controller;
```

```
import org.springframework.web.bind.annotation.GetMapping;
```

```
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
```

```
import com.cognizant.account.model.Account;
```

```
@RestController
```

```
public class AccountController {
```

```
    @GetMapping("/accounts/{number}")
```

```
    public Account getAccount(@PathVariable String number) {
```

```
        return new Account(
```

```
            number,
```

```
            "savings",
```

```
            234343
```

```
        );
```

```
    }
```

```
}
```